

Prediction Is Not Fact

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Boundaries and 'Not Sure'

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Build a Human Classifier

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Repeating Efficiently

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Check a boundary without changing the rule for Predictions and observations. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Predictions and observations. Proposed explanation: “Running the rule correctly proves the prediction correct”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Transfer the check to drying predictions

In a fictional paper test, Dry1 predicts slow for a moisture score of at least 7, quick otherwise. Allowed scores are whole numbers 1 through 10. The observed outcome is slow if drying time is greater than 10 minutes, quick otherwise. Records are P: score 6, time 12; Q: score 7, time 10; R: score 9, time 14; S: score 5, time missing. Record all predictions, observed categories and comparison statuses. Summarise matches out of assessed cases, name the unassessed count, and limit the conclusion.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

On Monday a team predicts 'rain on Tuesday'. On Tuesday they count a damp playground as success, although sprinklers ran. Write an observation rule that could have been agreed before Tuesday.

6. Solve the given case

A weather helper predicts tomorrow's rain.
Is this a fact about tomorrow? How can it be checked?
Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A model predicts rain tomorrow. Tomorrow is dry. Was the prediction an observation of tomorrow? Explain what new evidence settles the result.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 17 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Label the sorter's suggested bin as a prediction until a human confirms the final placement.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

On Monday a team predicts 'rain on Tuesday'. On Tuesday they count a damp playground as success, although sprinklers ran. Write an observation rule that could have been agreed before Tuesday.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Separate inputs from observed outcomes

For Ramp1 trial B, height is 6 cm and the later travel distance is 3 units. State the input, prediction, observation, observed category and comparison status. Explain why the prediction rule and outcome rule use different measurements.

My reasoning and evidence

13. W2 Check a boundary without changing the rule

Trial F has height 6 cm and verified travel distance exactly 4 units. Apply Ramp1 and the outcome rule. A partner says exactly 4 is near because far means greater than 4. Use the stated rule to resolve the disagreement.

My reasoning and evidence

14. W3 Keep incomplete evidence visible

D predicts near but has no distance record. E predicts far but its two distance records say 3 and 4 units. State each comparison status, request useful evidence, and explain why neither guessing nor a new trial can establish the old trial's outcome.

My reasoning and evidence

15. W4 Transfer the check to drying predictions

In a fictional paper test, Dry1 predicts slow for a moisture score of at least 7, quick otherwise. Allowed scores are whole numbers 1 through 10. The observed outcome is slow if drying time is greater than 10 minutes, quick otherwise. Records are P: score 6, time 12; Q: score 7, time 10; R: score 9, time 14; S: score 5, time missing. Record all predictions, observed categories and comparison statuses. Summarise matches out of assessed cases, name the unassessed count, and limit the conclusion.

My reasoning and evidence

16. W5 Correct a report without erasing evidence

After seeing B's observed near category, a learner replaces Ramp1's far prediction with near. They call the new entry a corrected observation and report a perfect result for A through C. Explain the two record-keeping errors and write a defensible replacement report.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Prediction Is Not Fact

A fictional drying trial uses a fixed rule and later measures what actually happened.

Model D predicts slow when moisture score is at least 5; quick otherwise.

Observation policy: slow if drying takes above 8 minutes; quick otherwise. Trial T has score 4 and measured time 9 minutes.

Prediction Is Not Fact

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Prediction input

T has moisture score 4.



2. Predict

4 is below 5, so D predicts quick.



3. Observe later

Measured time is 9 minutes; above 8 means observed slow.



4. Compare evidence

Prediction quick disagrees with observation slow.

CHANGE ONE THING

Keep score 4 but measure exactly 8 minutes.

Explain the changed result using the matching lesson.

Which field changes the observation label, and which field changes the model prediction?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: In sorter trials, record predicted categories before collecting reference evidence, then retain disagreements for investigation.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

In sorter trials, record predicted categories before collecting reference evidence, then retain disagreements for investigation.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional drying trial uses a fixed rule and later measures what actually happened.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Prediction quick disagrees with observation slow.

1. Predict
2. Compare evidence
3. Observe later
4. Prediction input

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare evidence Prediction quick disagrees with observation slow. The moisture score generates a prediction; it is not the observed drying time. The later measurement supplies separate evidence under the stated observation policy. Save both values and their units. Altering the recorded time to match D would destroy the evaluation rather than repair the model.

Practice 2: explain the deciding evidence

Changed-case practice: Keep score 4 but measure exactly 8 minutes. Which conclusion is supported by the supplied facts?

1. The observed label becomes quick and agrees with the prediction.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The observed label becomes quick and agrees with the prediction. Above 8 is strict: equality is quick under this policy.

Repair a missing building block

Review Boundaries and 'Not Sure' (ai-1/16). Goal: Identify a specific source of uncertainty and explain how it affects a prediction. Short practice: Which original block needs a better width before accepting or rejecting it? What should the reviewer do?. Return to this week and explain what you changed.

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated input format, selected rule and recorded output. Short practice: Trace 0 cm and explain why clarification is different from predicting short.. Return to this week and explain what you changed.

Review Repeating Efficiently (ai-1/11). Goal: Trace a loop by testing its condition before every repetition. Short practice: Trace a fresh start of 9. How many additions occur, and must the report equal 10?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
